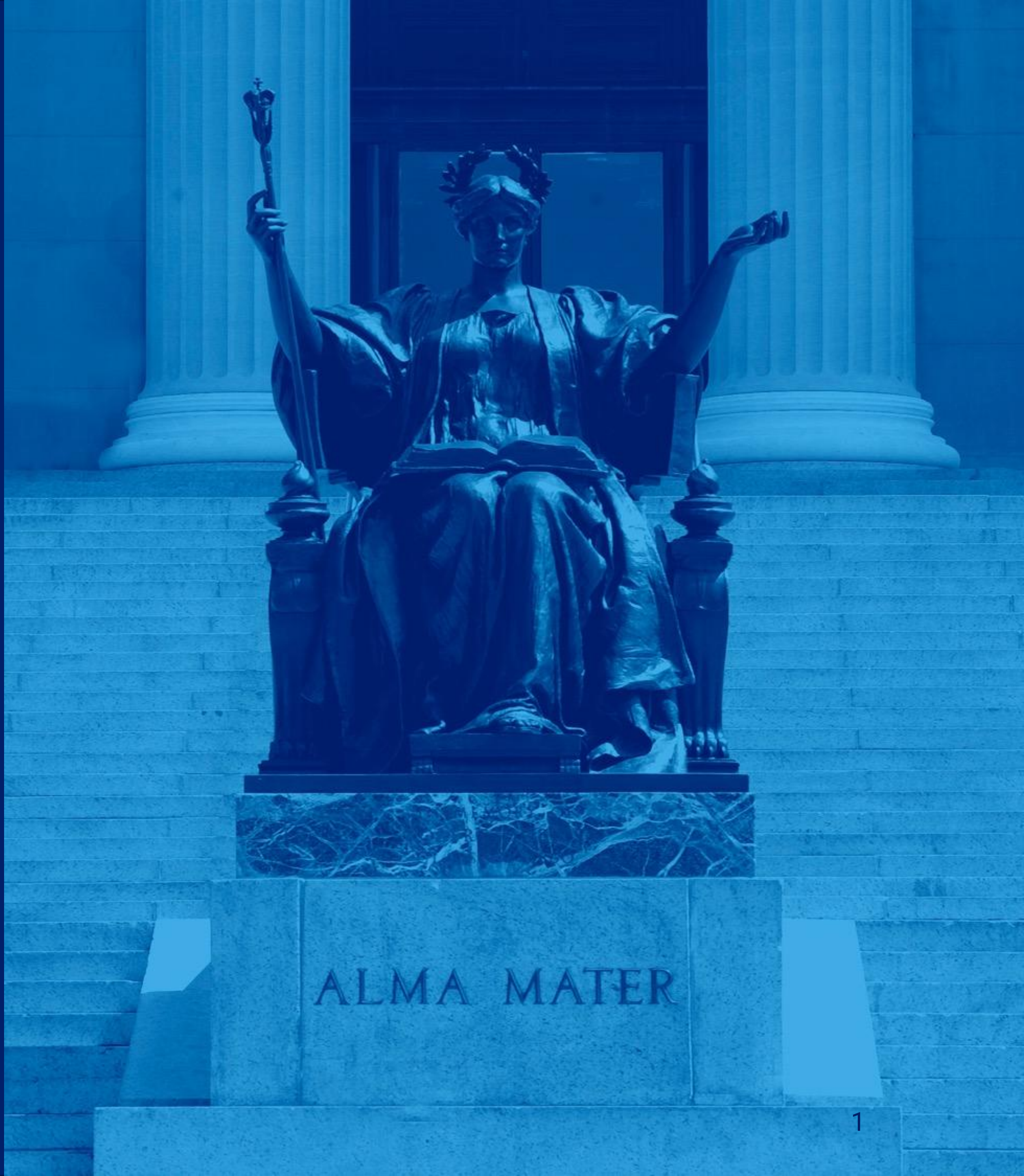
Columbia SIPA
School of International and Public Affairs

Generating Alpha Strategies using Machine Learning

April 2025



Project Overview

- Generate a successful alpha (beat the market) strategy through testing various technical indicators and fundamental metrics
- Which ML model provides the best result for ticker selection and forecasting price movements
- Does constant rebalancing yield significantly better performance assuming no commission fee

Investment Strategy

The strategy aims to exploit persistent signals in both fundamentals and price actions

- **Long:** Stocks trading above 200-day SMA, with ROE > 20% and Debt/Equity < 1
- **Short:** Stocks trading below 200-day SMA, with ROE < 10% and Debt/Equity > 2

Each Month:

- The top and bottom 10 (or less) stocks are selected based on ML model forecasts and the investment strategy
- The portfolio is equally weighted and rebalanced monthly
- Returns are compared to the S&P 500 Benchmark Performance

Data Pipeline

Data Collection

1. S&P 500 Tickers: The pipeline fetches the list of S&P 500 universe from Wikipedia
2. Price Data: Historical price data, ROE, DE are downloaded using finance
3. S&P 500 Index performance is downloaded for benchmark purposes

Data Organization (Sample)

- Training Period: 2017-01-01 to 2020-12-31 (four year)
- Validation Period: 2020-12-31 to 2021-12-31 (one year)
- Testing Period: 2021-12-31 to 2022-12-31 (one year)

Feature Engineering (The “X”)

Technical Indicators

- Moving Averages
- SMA ratios
- RSI (Relative Strength Index)
- MACD (Moving Average Convergence Divergence)
- Volatility metrics
- Bollinger Band width

ARIMA-based Features

- Predictions for 5, 10 and 21-day forward returns
- ARIMA model parameters (ARI, MA1, constant)
- Model quality metrics (AIC, volatility)

Fundamental Data

- ROE (return on Equity)
- Debt-to-equity ratio (measure of leverage)

Market Regime Filter

- Binary indicator of bull/bear market based on the 200-day S&P 500 moving averages

Target Variable (The "Y")

1. Calculate forward returns over the next 21 trading days
2. Convert this continuous return into three classes:
   - -1: Poor (0 - 0.25)
   - 0: Neutral (0.25 - 0.75)
   - 1: Outperforming (0.75 - 1)

Model Training

Logistic Regression

- Each feature is multiplied by a coefficient (weight), that represents the change in log-probability of a stock belonging to a particular class relative to the reference class
- The weighted features are summed together with a bias term
- A multinomial logistic regression then calculates 3 different linear combinations of input features which are the 3 classes of prediction

K-Nearest Neighbours

- Each stock is represented as a point in an n-dimensional space, where n is the number of features (technical indicators, ARIMA, fundamental etc.)
- When making a prediction, the model calculates the distance between new point and all training points and identifies k = 5 closest training examples
- The predicted class is determined by majority vote among the k neighbours

Model Training Cont.

LSTM Neural Network

1. **Sequence Creation**
   - Transforms time series data into sequences of 20 consecutive days for each stock
   - Each sequence contains 20 days of feature data, and the target is return label for the day following the sequence
2. **Architecture**
   - First LSTM layer with 50 units, returning seuqnces for the next layer
   - Dropout (0.2) to prevent overfitting
   - Second LSTM layer with 50 units
   - Another Dropout layer
   - Final Dense layer with 3 units (one for each class) and softmax activation
3. **Compilation**
   - Adam optimizer with learning rate of 0.001
   - Categorical cross-entropy loss
   - Accuracy metric for evaluation
4. **Training**
   - Trained for 10 epochs with batch size 32, using the validation set to monitor performance

An **ensemble** is created by averaging the three individual model predictions for labels (-1,0,1)

Portfolio Construction

1. The prediction output is then used to build a trading portfolio filtered by the investment strategy

- **Long:** Stocks trading above 200-day SMA, with ROE > 20% and Debt/Equity < 1
- **Short:** Stocks trading below 200-day SMA, with ROE < 10% and Debt/Equity > 2

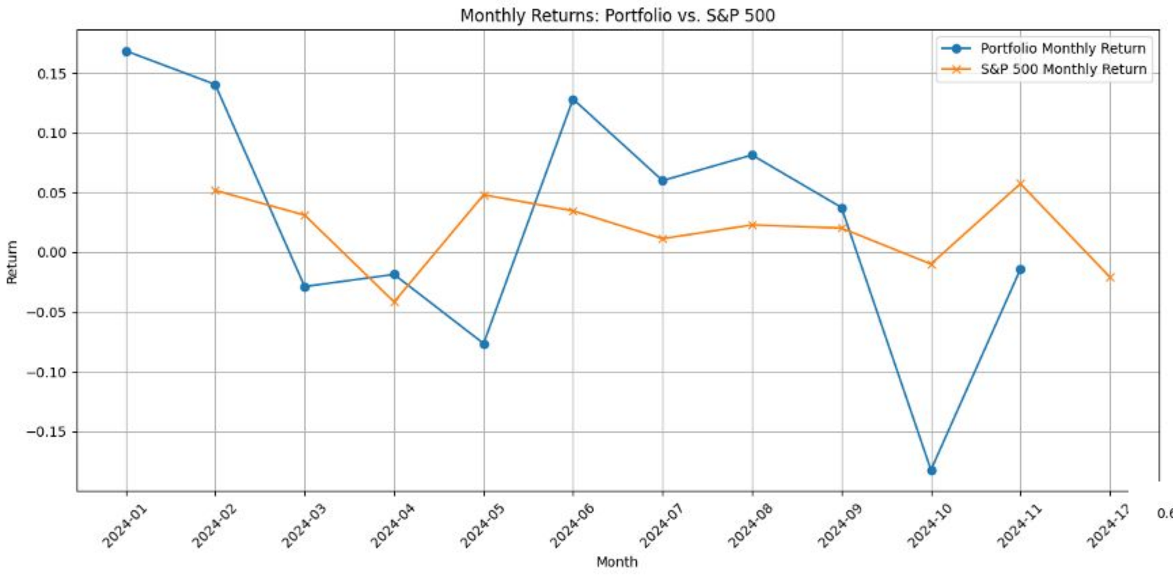
2. Within the filtered sets, tickers are sorted by the ensemble predictions
3. The top and bottom < 10 stocks for long and short positions are chosen for the final portfolio

Backtesting and Evaluation

- The ensemble method produced more stable returns while the LSTM model had more extreme performances

- In times of market drawdowns, the ensemble method were able to minimize losses by identifying the market regime and short (bull vs bear), while the LSTM-only model had years with very poor returns

Appendices (2024 Ensemble)



Benchmark Performance Summary:

Total Return: 0.2454

Annualized Return: 0.2476

Annualized Volatility: 0.1267

Sharpe Ratio: 1.7963

Maximum Drawdown: -0.1014

Portfolio Performance Summary:

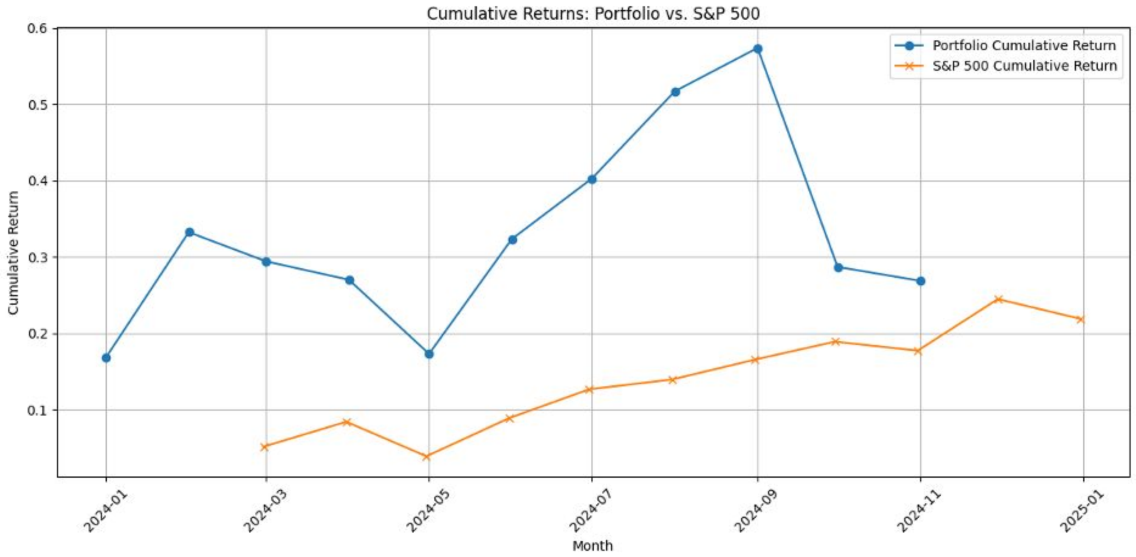
Total Return: 0.2688

Annualized Return: 0.2965

Annualized Volatility: 0.3609

Sharpe Ratio: 0.7663

Maximum Drawdown: -0.1937



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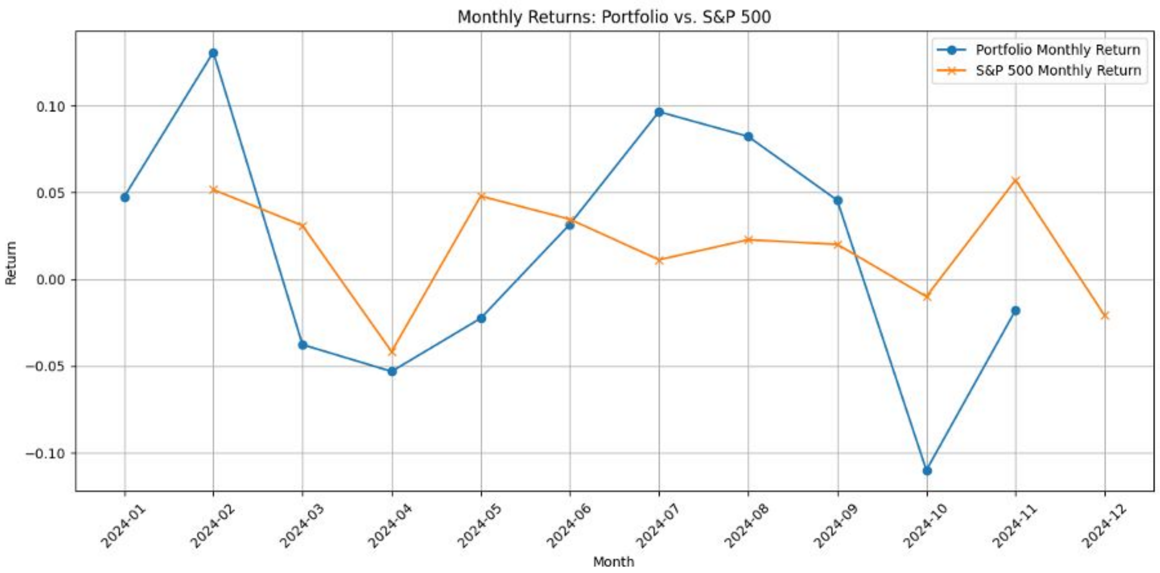


Benchmark Performance Summary:

Total Return: 0.2454

Annualized Return: 0.2476

Annualized Volatility: 0.1267

Sharpe Ratio: 1.7963

Maximum Drawdown: -0.1014

Portfolio Performance Summary:

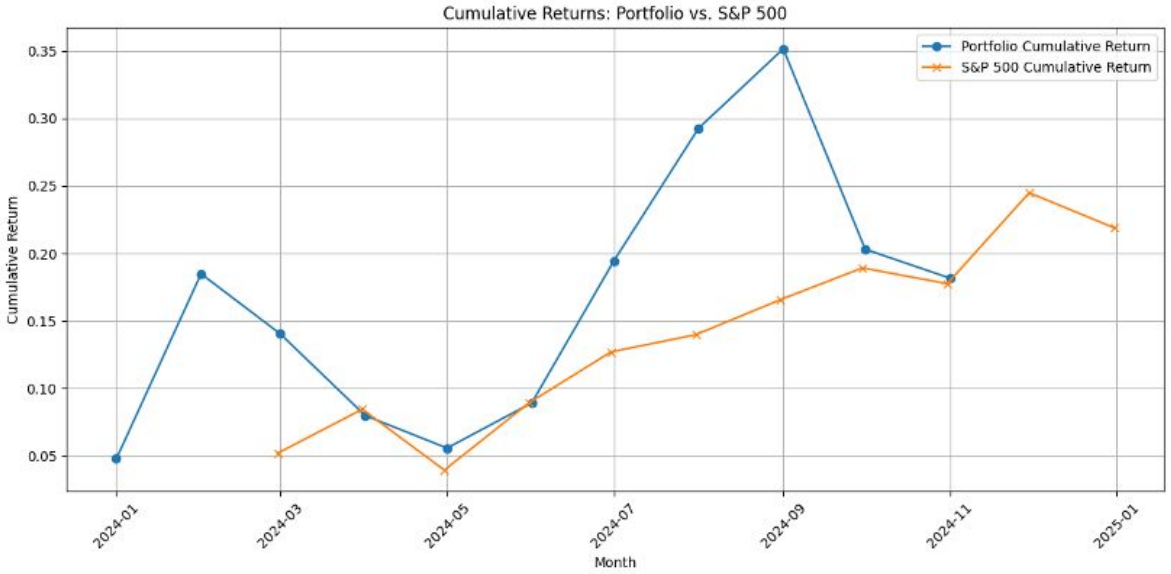
Total Return: 0.1816

Annualized Return: 0.1996

Annualized Volatility: 0.2506

Sharpe Ratio: 0.7167

Maximum Drawdown: -0.1258



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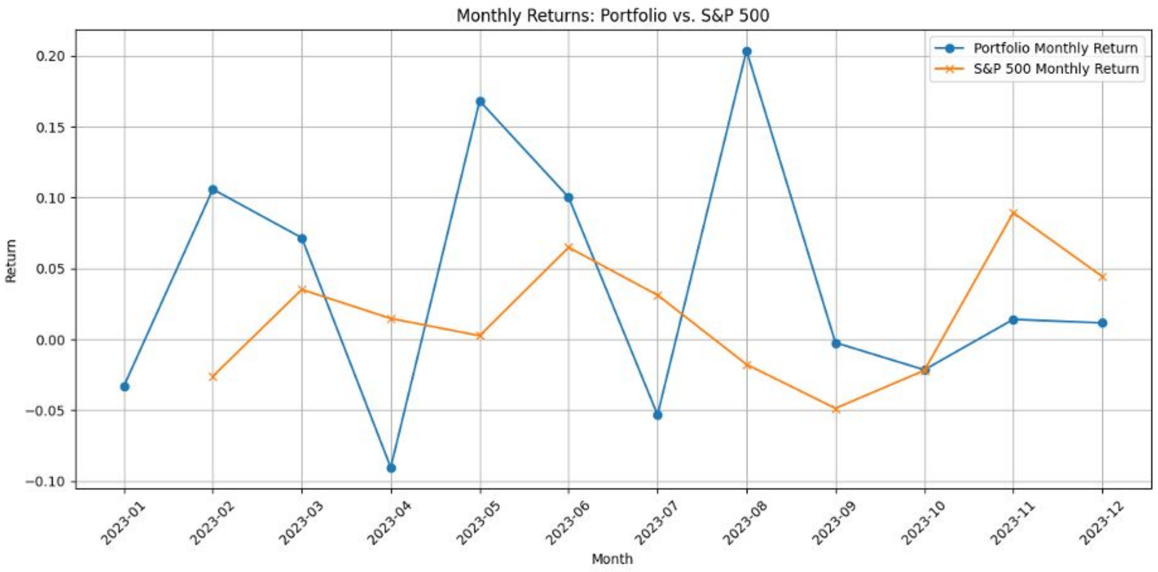


Benchmark Performance Summary:

Total Return: 0.2473

Annualized Return: 0.2506

Annualized Volatility: 0.1311

Sharpe Ratio: 1.7591

Maximum Drawdown: -0.1233

Portfolio Performance Summary:

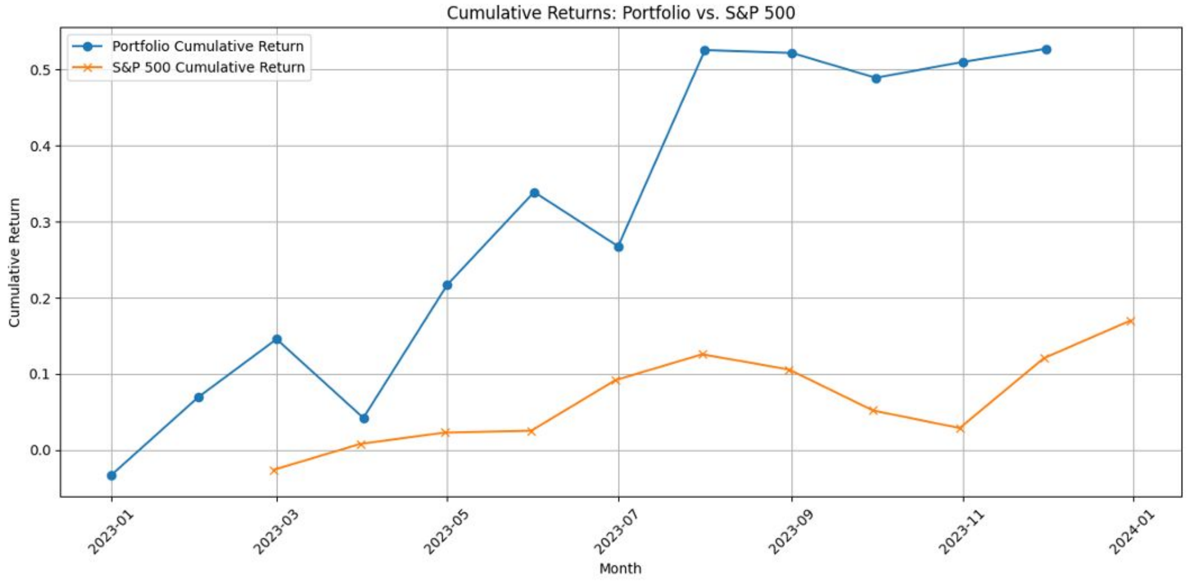
Total Return: 0.3533

Annualized Return: 0.3533

Annualized Volatility: 0.3040

Sharpe Ratio: 1.0964

Maximum Drawdown: -0.1182



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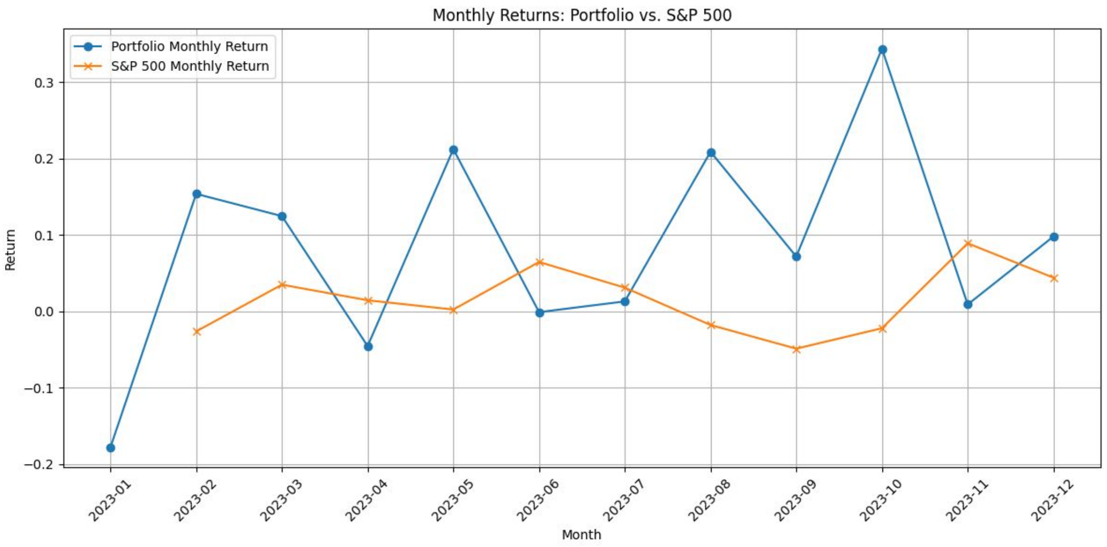


Benchmark Performance Summary:

Total Return: 0.2473

Annualized Return: 0.2506

Annualized Volatility: 0.1311

Sharpe Ratio: 1.7591

Maximum Drawdown: -0.1233

Portfolio Performance Summary:

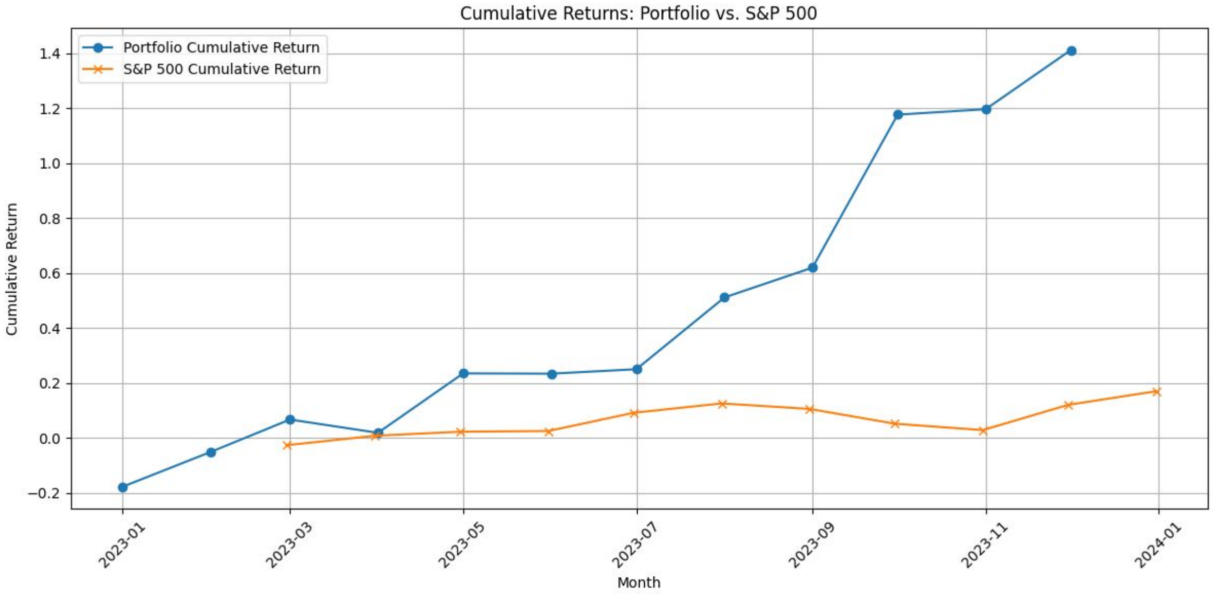
Total Return: 1.4123

Annualized Return: 1.4123

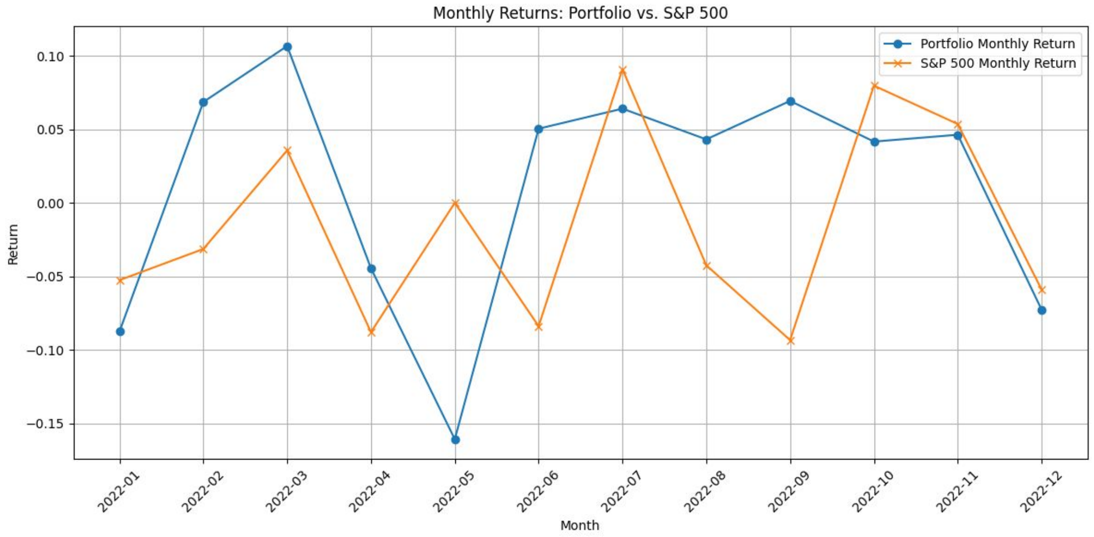
Annualized Volatility: 0.4781

Sharpe Ratio: 2.9121

Maximum Drawdown: -0.0453



Appendices (2022 Ensemble)



Benchmark Performance Summary:

Total Return: -0.1944

Annualized Return: -0.1951

Annualized Volatility: 0.2417

Sharpe Ratio: -0.8899

Maximum Drawdown: -0.2559

Portfolio Performance Summary:

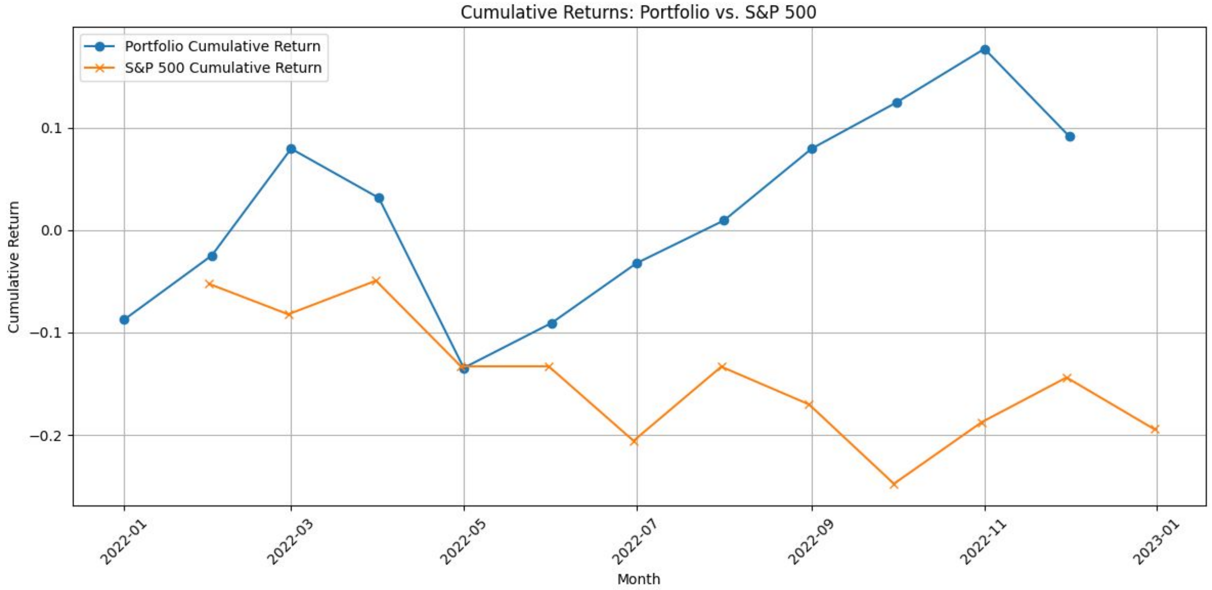
Total Return: 0.0913

Annualized Return: 0.0913

Annualized Volatility: 0.2820

Sharpe Ratio: 0.2529

Maximum Drawdown: -0.1982



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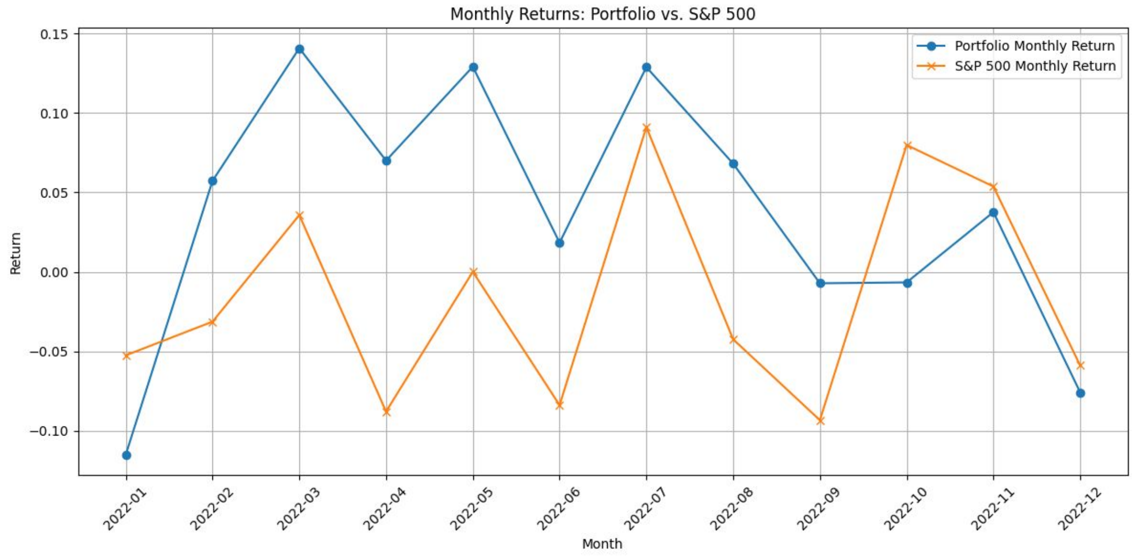


Benchmark Performance Summary:

Total Return: -0.1944

Annualized Return: -0.1951

Annualized Volatility: 0.2417

Sharpe Ratio: -0.8899

Maximum Drawdown: -0.2559

Portfolio Performance Summary:

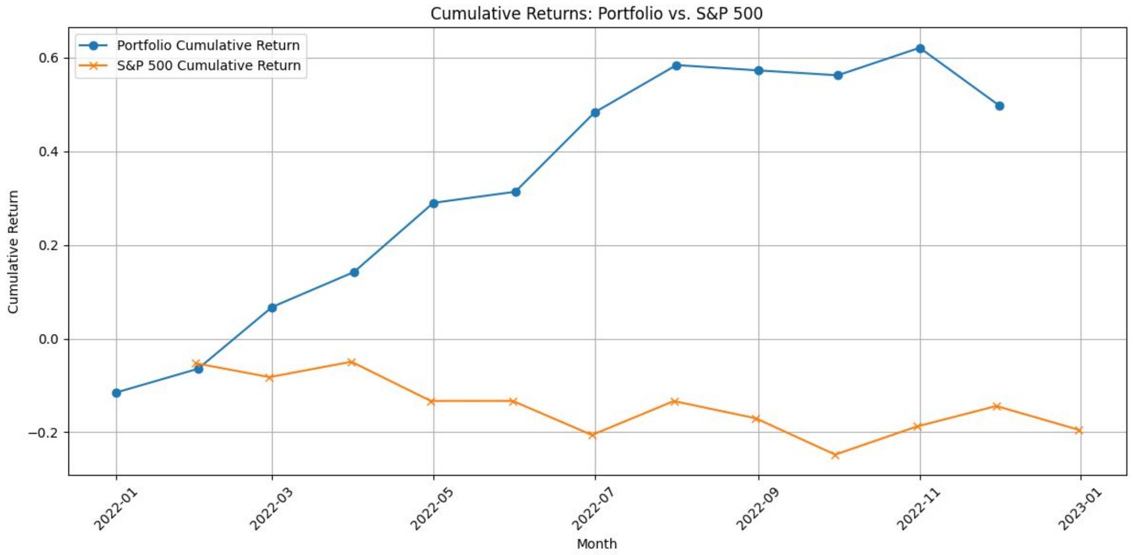
Total Return: 0.4977

Annualized Return: 0.4977

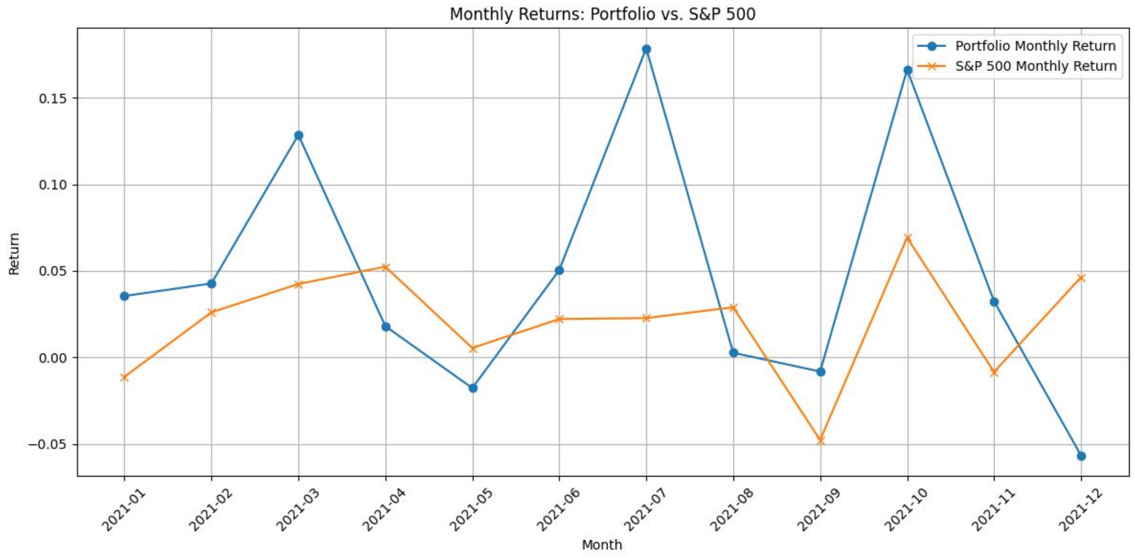
Annualized Volatility: 0.2769

Sharpe Ratio: 1.7250

Maximum Drawdown: -0.0759



Appendices (2021 Ensemble)



Benchmark Performance Summary:

Total Return: 0.2723

Annualized Return: 0.2735

Annualized Volatility: 0.1312

Sharpe Ratio: 1.9319

Maximum Drawdown: -0.0630

Portfolio Performance Summary:

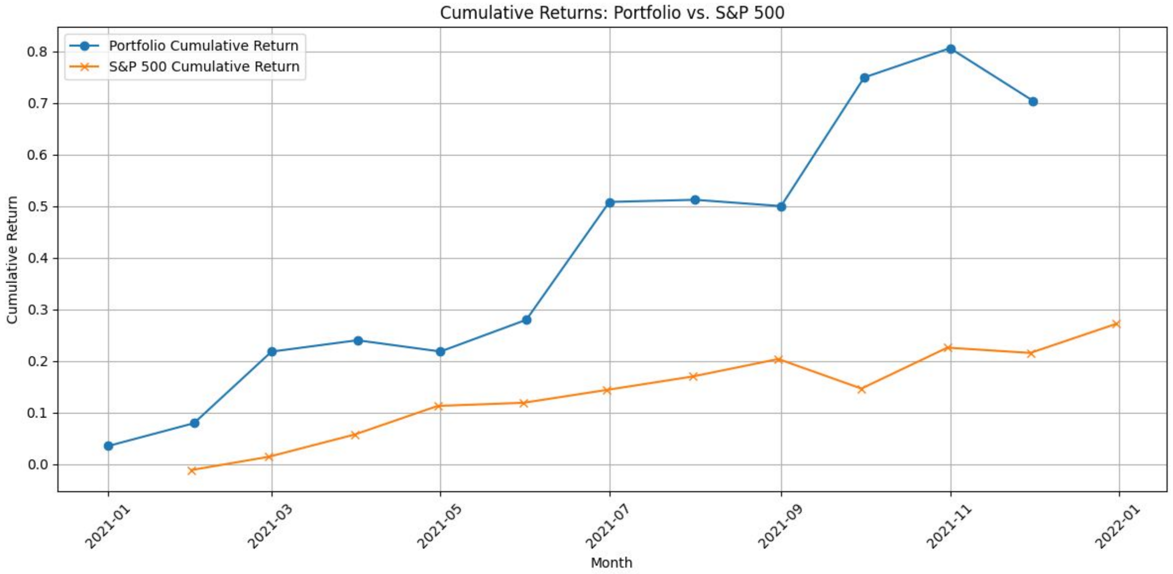
Total Return: 0.7044

Annualized Return: 0.7044

Annualized Volatility: 0.2541

Sharpe Ratio: 2.6935

Maximum Drawdown: -0.0565



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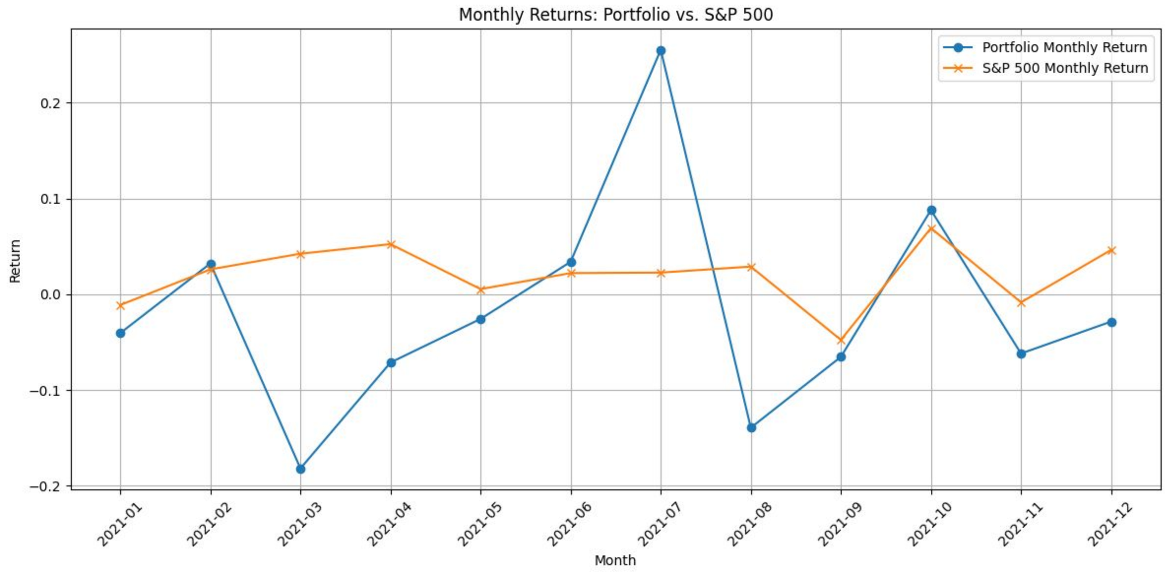

Benchmark Performance Summary:

Total Return: 0.2473

Annualized Return: 0.2506

Annualized Volatility: 0.1311

Sharpe Ratio: 1.7591

Maximum Drawdown: -0.1233

Portfolio Performance Summary:

Total Return: -0.2404

Annualized Return: -0.2404

Annualized Volatility: 0.3914

Sharpe Ratio: -0.6654

Maximum Drawdown: -0.2595